Hierarchical percolation of travel time reveals spatial structure in the Roman road network

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Abstract

We apply hierarchical agglomerative clustering (HAC) on travel-time distances – under single linkage, mathematically equivalent to a percolation sweep over the network's own edges – to the Roman road network, with the aim of recovering spatial units of interest directly from network connectivity rather than from imposed administrative boundaries. Before turning to our own network, we calibrate the method on the exact network and travel-time model used in a recent large-scale reconstruction of the Roman road system, and show that it recovers that network's Roman provinces with an Adjusted Rand Index (ARI) of 0.70 (bootstrap 95% CI [0.68, 0.71]) at a threshold of ∼77 hours (∼3.2 days) of travel – confirming that the method can recover a known administrative partition when one exists in the data. We then apply the same percolation-equivalent clustering to our own, independently constructed network of the Roman road system, where the goal is not to reproduce this calibration result but to characterize the emergent spatial units the network itself suggests. A first, coarse benchmark against the Roman provinces on this network gives a markedly weaker alignment (ARI ≈ 0.43) with a less sharply defined optimal threshold, which we read not as a replication failure but as a signal that provincial administration is only one of several candidate organizing principles the clustering may be picking up – motivating a closer characterization of these units, and a comparison against spatial/temporal null models, as the next step of this work.

1 Introduction

Introduction placeholder – literature review and broader framing to be completed. The two questions below should survive into whatever full introduction is eventually written; everything else here is a placeholder.

This paper is organized around two questions, both posed on the network built from the Itiner-e Roman road data [de Soto et al., 2025]:

1. **Can hierarchical agglomerative clustering (HAC) on travel-time distances identify spatial units that emerge from the road network itself?** We use percolation-equivalent HAC (Section 2.2) to cluster the network at every travel-time threshold, calibrate the approach against a known partition (Section 3.1), and then apply it to our own network to see what clusters it produces on its own terms (Section 3.2).

2. **Are the spatial units that emerge significant, or are they artefacts of the network's own construction?** A cluster can look like a coherent region without reflecting anything beyond the geometry and density of the digitized road network it was built from. Answering this requires comparing the clusters obtained on the real network against the same procedure run on spatial and temporal null models of that network – i.e. networks that share its nodes and broad connectivity but not whatever structure actually produces

the clustering. This comparison, against the null models in `results/Null-models/`, is the next stage of this work and is not yet part of this manuscript (see Section 3.2 and the Discussion).

2 Methods

2.1 Data

The road network is built from the digitized linework released as the Itiner-e dataset [de Soto et al., 2025], cleaned following the same general procedure as Pažout et al. [2026]: near-miss junctions and duplicate endpoints are snapped together, the linework is noded at every intersection, and the resulting degree-two chains are simplified away, giving an undirected planar graph whose nodes are road intersections and dead ends and whose edges are the road segments between them. We use two variants of this network. Our own, independently cleaned version has 8,460 nodes and 12,849 edges across 12 connected components, with each edge weighted by travel time at a constant walking speed of 2 km/h. For calibration, we separately use the exact endpoint-noded network and travel-time model of Pažout et al. [2026] themselves – 10,506 nodes and 14,770 edges across 13 connected components, including two artificial crossings at the Bosporus and the Dardanelles connecting the European and Asian landmasses, with travel time estimated per edge from Tobler's hiking function [Tobler, 1993] applied to slope and geodesic length rather than a flat walking speed. Both networks are overlaid with the same province boundaries: the Ancient World Mapping Center's "Roman Empire AD 200" polygons, corrected by A. Pažout [Ancient World Mapping Center, 2023], approximate and clipped to modern coastlines, comprising 57 provinces with Italy further subdivided into its constituent regiones for finer resolution on the Italian peninsula – a division not available for other provinces – alongside a coarser, 47-unit partition that merges the regiones back into a single Italia for comparison. Each road edge is assigned to the province containing its midpoint, falling back to the province with the longest geometric overlap where the midpoint falls outside every polygon; nodes inherit the province of their incident within-province edges, and edges that cross a provincial boundary or cannot be assigned to any province are excluded from the alignment scoring in Section 3.

2.2 Hierarchical agglomerative clustering and percolation

Hierarchical clustering under single linkage has a direct reading on a weighted network: merging two clusters at the weight of the lightest edge joining them means that, at any linkage height t, the clusters are exactly the network's connected components once every edge heavier than t has been removed. Raising t and cutting the dendrogram at each height is therefore the same thing as percolating the network itself, component by component, rather than clustering points by a chosen distance metric. This is the basis of Arcaute et al. [2016], who percolate street intersections in Britain by Euclidean distance to recover its regional and urban hierarchy, and of Molinero et al. [2017], who percolate the roads themselves using the relative angle between consecutive segments as the occupation cost. We apply the same principle to travel time: within each connected component we compute pairwise travel-time distances between nodes and use single-linkage HAC, so that cutting the dendrogram at threshold t recovers exactly the network's own travel-time percolation clusters at that threshold. Our calibration network instead uses average linkage, inherited from work that predates this percolation framing and kept here as a deliberate point of comparison; it does not share the guarantee above, so its cluster counts and province-alignment scores in Section 3 are not directly comparable to those of our own network.

Sweeping t across every merge height gives, at each threshold, the number of clusters and the size of the largest one relative to the network (the percolation curve), to which we fit a power law over its emergent region. At each threshold t, we score the resulting clusters against

the province labels of nodes with a known province assignment using the Adjusted Rand Index [ARI; Hubert and Arabie, 1985], and identify the travel-time threshold at which this alignment is strongest.

2.3 Entropy of cluster sizes

As clusters merge during the percolation sweep, the distribution of cluster sizes changes shape. Following Molinero et al. [2017], we plan to track the Shannon entropy of this distribution as a function of threshold: local peaks in that entropy curve mark the points where the system passes between qualitatively different stages of integration, giving a more principled way to locate transitions in the percolation process than reading them off the cluster-count curve by eye. This analysis has not yet been carried out on either network; it is introduced here as the next planned step.

2.4 Comparison with spatial null models

To test whether the clusters found above reflect more than the geometry of the network's node positions, we compare them against spatial null models built on the same nodes but not the same roads: a nearest-neighbour network and a Delaunay-triangulation network, each reinforced with a minimum-spanning-tree backbone so that every node remains reachable. Figure 1 and Figure 2 show the same percolation procedure applied to these two null networks. In both, the number of clusters falls and the largest cluster grows smoothly and gradually with threshold, without the plateaus visible in the real network's percolation curve (Figure 3). A full comparison, scoring the real network's clusters against these null models threshold by threshold, is still in progress, and will determine whether the stages found in the real network are a genuine signal or an artefact of its geometry alone.

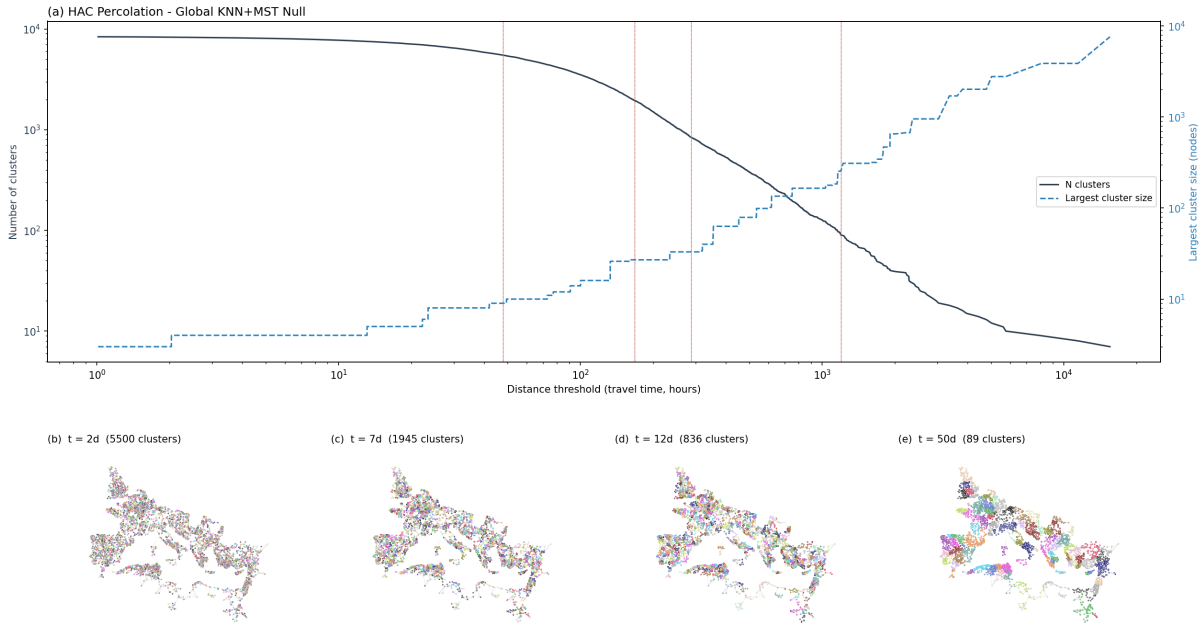


Figure 1: Percolation curve and snapshot cluster maps for a nearest-neighbour spatial null network sharing the real network's nodes.

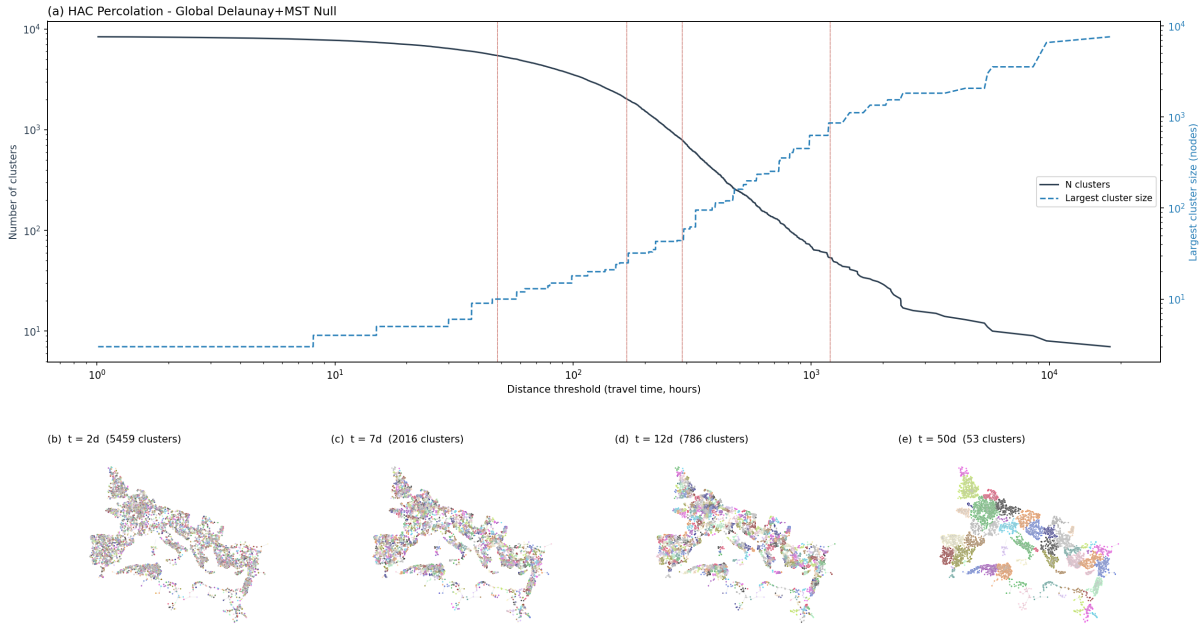


Figure 2: Percolation curve and snapshot cluster maps for a Delaunay-triangulation spatial null network sharing the real network's nodes.

3 Results

3.1 Identifying spatial units using HAC

Before asking what spatial units our own network suggests, we first check that the method is capable of recovering a known partition at all, on a network and travel-time model we did not build ourselves. Figure 3 shows the percolation curve and snapshot cluster maps for the calibration network of Pažout et al. [2026], with Tobler-function travel times and AWMC provinces (10,430 of 10,506 nodes have a known province).

Table 1 reports ARI and NMI at six representative travel-time snapshots. Alignment with the provinces rises from essentially zero at 6–12 hours, peaks around 1–3 days, and falls off again as clusters grow beyond provincial scale.

Table 1: Province alignment at fixed travel-time snapshots, calibration network (Pažout et al. [2026], Tobler-function travel time; $n = 10{,}430$ nodes with known province).

Snapshot	Clusters	ARI	NMI
6 h	2,809	0.039	0.640
12 h	1,376	0.092	0.683
24 h	514	0.201	0.734
2 d	154	0.516	0.803
7 d	22	0.493	0.778
12 d	15	0.308	0.689

An exhaustive search over every merge height finds a single, sharp peak at $t = 77.1$ hours ($\sim$3.2 days), with ARI $= 0.695$ (95% bootstrap CI $[0.681, 0.710]$). The peak is stable: every bootstrap resample of the node set selects the same 77-hour merge height as optimal. At this threshold the network's emergent regions correspond closely to the Roman provincial map.

Figure 4 shows ARI as a continuous function of threshold rather than at the six fixed snapshots of Table 1, making the sharpness of the 77-hour peak visible directly, and Figure 5 breaks

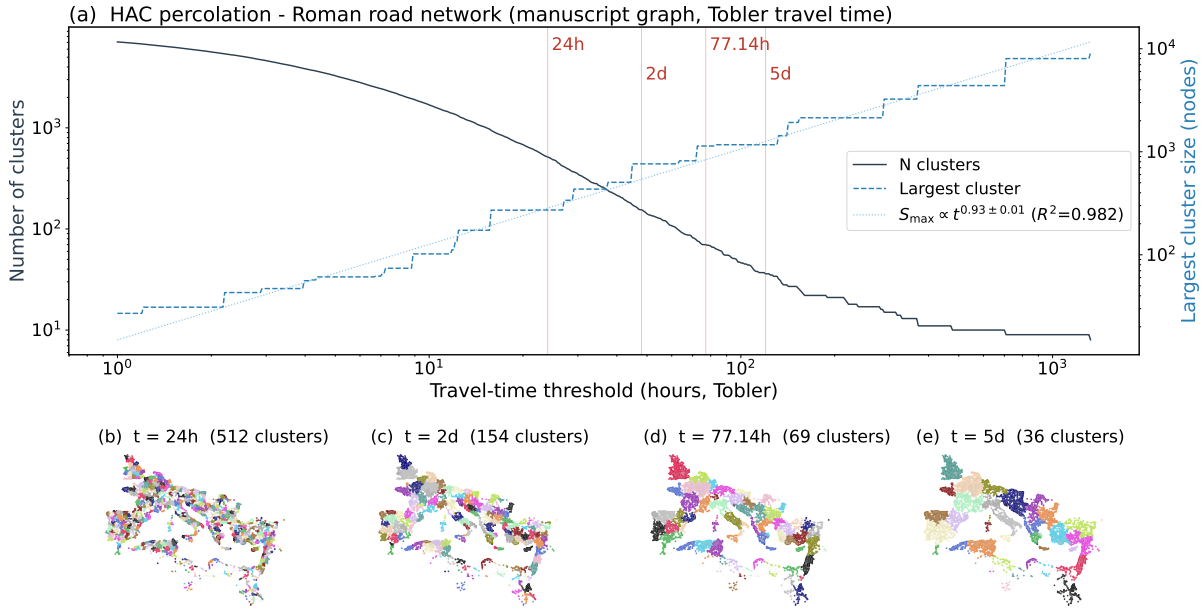


Figure 3: Percolation curve and snapshot cluster maps for the calibration network.

that peak down province by province: for each province, how well its nodes stay together in a single dominant cluster (recall) and how much of that cluster is specific to it (precision), combined into an F1 score. Small, geographically isolated provinces (Britannia, Cyprus, Creta, Sicilia, Sardinia, Corsica) are recovered almost perfectly (F1 $\approx$ 1), as expected since islands and peripheral regions have little opportunity to merge with neighbours at this threshold; the worst-recovered provinces are the small Alpine provinces (Alpes Graiae, Cottiae, Poeninae, Maritimae) and Germania Superior, which sit on travel corridors between larger, better-connected neighbours and are absorbed into them rather than forming distinct clusters of their own. This province-by-province view is what we will be looking for an equivalent of in Section 3.2, once our own network's clusters have had the same closer look.

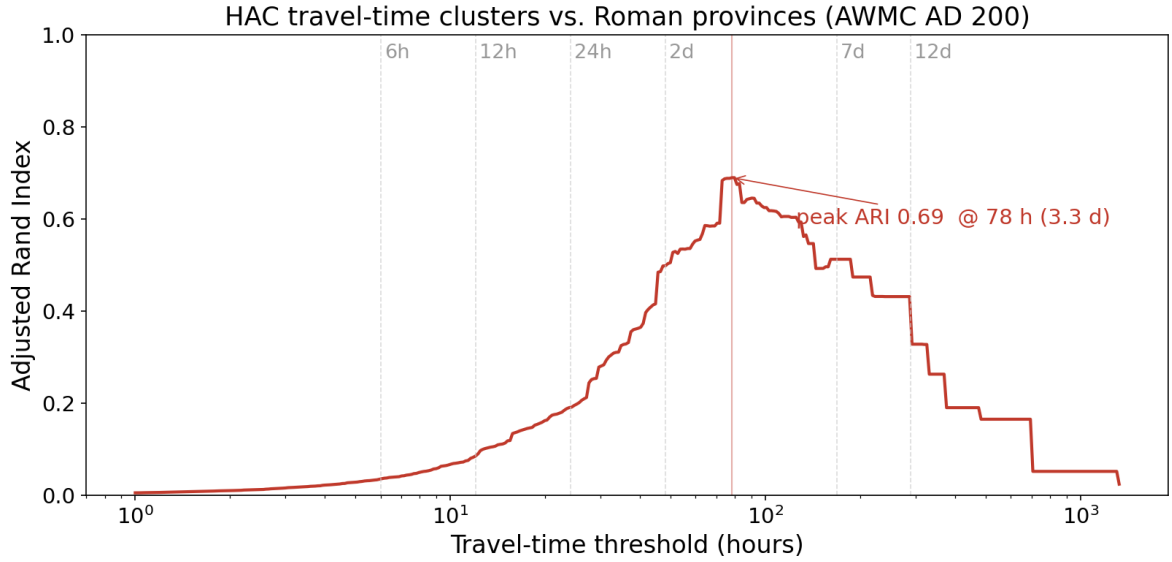


Figure 4: ARI between HAC clusters and Roman provinces as a continuous function of travel-time threshold, calibration network.

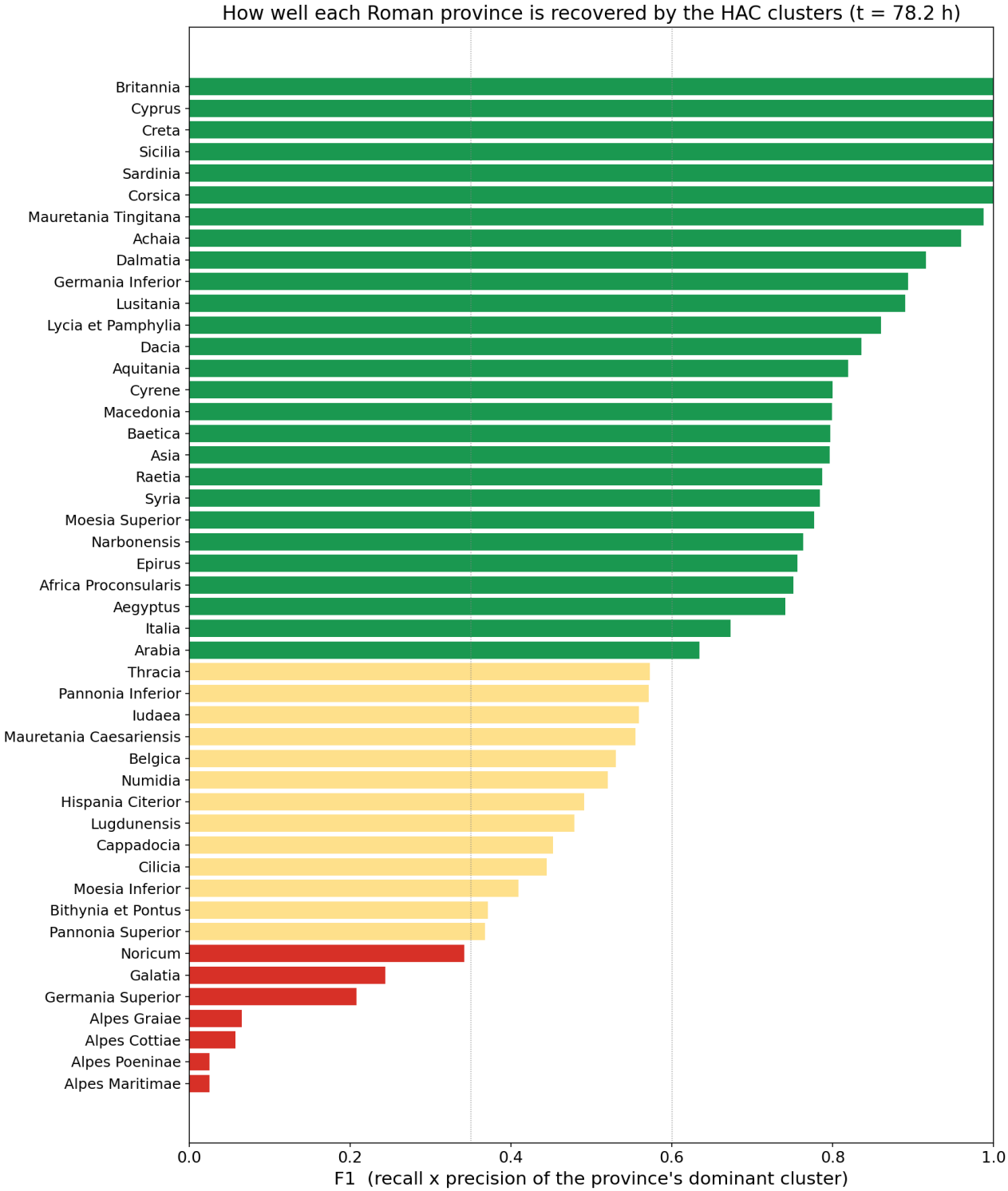


Figure 5: Per-province F1 (recall × precision of each province's dominant cluster) at the peak-alignment threshold ($t = 78.2$h), calibration network.

3.2 Applying HAC to the itinereX network

Having confirmed in Section 3.1 that percolation-equivalent HAC can recover a known administrative partition when one is present, we turn to the first question posed in Section 1: what spatial units does the itinereX network itself suggest, under its own independently built topology and a simple constant-speed (2 km/h) travel-time model? This is not a replication of Section 3.1 – we are not trying to reproduce its ARI on a different network, and treating the comparison that way would misread what each run is for. Section 3.1 establishes that the method works; this section applies it, for the first time, to a network nobody has clustered before, with the goal of characterizing the clusters themselves as candidate spatial units, of which provincial alignment is only one possible interpretation among several worth testing. The second question – whether those units are significant, or artefacts of this particular network's construction – is addressed only partially here, and is taken up properly in the paragraph after next.

As a first, coarse orientation, we benchmarked the resulting clusters against the Roman provinces using the same ARI/NMI procedure as Section 3.1: an exhaustive search over every merge height gives a best alignment of ARI $= 0.433$ (NMI $= 0.763$, 291 clusters) at $t \approx 31.8$ hours, with a second, comparably good local optimum around 25 hours rather than the single sharp peak seen in Section 3.1 (95% bootstrap CI on ARI $[0.426, 0.448]$; on the threshold, $[24.8, 31.8]$ hours). We do not read this as a weaker version of Section 3.1's result – on a network built under a different travel-time model, with its own topology, there is no reason to expect the same number – but as a preliminary signal that provincial administration is detectable in, without fully explaining, the clusters this network produces. We have deliberately not built a figure or table around this number here: a single province-alignment statistic does not do justice to what the clusters at each threshold actually look like, and a closer, better-presented look at them is needed before drawing conclusions from it.

The more interesting question is not why the ARI is lower here, but what these clusters represent. That is exactly the second question raised in Section 1, and is taken up in Sections 2.3 and 2.4.

4 Discussion

Discussion placeholder – to be completed. Note for later: the calibration run (Section 3.1) establishes the method works; the open question for network1-single is what its emergent clusters represent, which needs (i) a proper figure/table once the clusters have had a closer look, and (ii) comparison against the null models in `results/Null-models/` to separate a genuine topological signal from an artefact of the constant-speed travel-time model.

Data accessibility

All code and data underlying this analysis are available in the accompanying repository, under `results/HAC/network1-single/` (primary analysis) and `results/HAC/network0-average/` (calibration analysis).

Authors' contributions

To be completed.

Competing interests

The authors declare no competing interests.

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To be completed.

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